

1. Administrative & Study Identification

Full Study Title: A Multicentre, Randomised, Double-blind, Parallel Group, Placebo-controlled, Time-to-first Asthma Exacerbation Phase III Efficacy and Safety Study of Benralizumab in Paediatric Patients With Severe Eosinophilic Asthma (DOMINICA)
Short Title/Acronym: DOMINICA **Protocol Number:** D3250C00024 **NCT Number:** NCT05692180 **Posting ID:** 385554746964797933 **Trial Status:** RECRUITING **Sponsor/Company Name:** AstraZeneca (Collaborator: PAREXEL) **Investigational Product:** Benralizumab (subcutaneous injection) **Trade Name:** Fasenra **ATC Code:** R03DX10 **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of benralizumab administered subcutaneously in paediatric patients (≥ 6 to < 18 years of age) with severe eosinophilic asthma, specifically assessing the time-to-first asthma exacerbation. **Secondary Objectives:** To evaluate the effectiveness of benralizumab in improving asthma control (via ACQ-IA), reducing asthma symptom scores, and decreasing rescue medication use during the double-blind treatment period.

2.2 Background & Rationale Asthma is a chronic respiratory disease manifested as difficulty breathing due to the narrowing of bronchial passageways. An increase in eosinophils (a type of white blood cell) is strongly related to severe asthma pathology. Benralizumab works by eliminating these eosinophils, which can help control asthma symptoms, a mechanism that has already been successfully demonstrated in adult patients with severe asthma. This study is conducted to determine if these same efficacy and safety benefits extend to paediatric patients aged 6 to < 18 years with uncontrolled asthma receiving high-dose ICS and at least one additional controller.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Parallel Group **Blinding:** Double-blind (during the initial efficacy period) followed by an Open-Label Extension (OLE) **Randomization:** 1:1 to receive benralizumab or placebo

3.2 Planned Enrollment & Duration Target **N**: 200 participants
Number of Sites: Multicentre (Global) **Estimated Study Start:** 05-Apr-2023 **Estimated Primary Completion:** 05-May-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients aged ≥ 6 to < 18 years old capable of giving assent (with caregiver written informed consent). 2. Physician-diagnosed severe eosinophilic asthma for at least 12 months prior to Visit 1, and managed by the clinical site/site network for ≥ 6 months prior to Visit 1. 3. Exacerbation history defined as a requirement for systemic corticosteroids and/or hospitalisation within 12 months prior to Visit 1; OR 2 asthma exacerbations per year within the 2 years prior to Visit 1 AND currently on stable maintenance oral corticosteroids (OCS) used for at least 3 months OR at least one of the 2 exacerbations in the prior year resulted in hospitalisation. 4. Patients on well-documented, stable treatment for asthma with high-dose inhaled corticosteroids (ICS) and at least one additional controller medication (e.g., LABA, LTRA, LAMA, or theophylline) for at least 6 months prior to Visit 1. 5. Eosinophilic airway inflammation indicated by peripheral blood eosinophil count of ≥ 300 cells/ μL during screening, OR a blood eosinophil count of 150 to 299 cells/ μL with documented elevated eosinophils in BAL, sputum, or bronchial biopsy within the prior 2 years. 6. $\geq 70\%$ compliance with maintenance asthma medication during the screening period; and at least 70% daily PASO or Asthma Daily Diary completion during screening, with at least 50% completion in the 14-day period prior to randomisation. 7. Pre-BD FEV1 $\leq 95\%$ of predicted normal or pre-BD FEV1/FVC ratio < 0.85 at Visit 1 (patients with $\geq 25\%$ increase in pre-BD FEV1 during screening will fail). 8. ACQ-IA ≥ 1.5 with no meaningful improvement (change ≤ -0.5) between screening and Visit 2a. 9. Body weight ≥ 15 kg. Females of childbearing potential (FOCBP) who are sexually active must use a highly effective method of contraception.

4.2 Exclusion Criteria 1. Clinically important pulmonary disease other than asthma, or systemic diseases associated with elevated peripheral eosinophil counts. 2. Life-threatening asthma or an asthma exacerbation requiring the use of systemic corticosteroids or an increase in maintenance OCS within 2 weeks prior to Visit 2a, or acute respiratory infection requiring antibiotics/antivirals within 2 weeks prior to Visit 2b. 3. History of anaphylaxis to any biologic therapy or known hypersensitivity to benralizumab/excipients. 4. Currently pregnant, breastfeeding, or lactating females. 5. Current malignancy, or history of malignancy. 6. A helminth parasitic infection. 7. Use of immunosuppressive medication. 8. Receipt of immunoglobulin or blood products within 30 days prior to Visit 1. 9. Receipt of any marketed or investigational biologic within 5 half-lives prior to Visit 1, or previously received benralizumab (MEDI-563).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Benralizumab active solution or matching placebo. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Minimum 16 weeks in the double-blind (DB) treatment period, continuing until an exacerbation occurs or required events are observed. All exacerbating patients are offered to enter the Open-Label Extension (OLE) period (at least 48 weeks for patients 12 to <18 years, and at least 104 weeks for patients 6 to <12 years).

5.2 Concomitant & Prohibited Therapy Permitted: Maintenance asthma therapies (high-dose ICS plus controller medications like LABA, LTRA, LAMA). **Prohibited:** Receipt of any marketed or investigational biologic within 5 half-lives prior to Visit 1, immunosuppressive medication, immunoglobulins or blood products within 30 days.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 2 months (min. 4 weeks)	Informed Consent/Assent, Medical History, Eosinophil Labs, Medication Compliance Check
Baseline (Visit 2)	Split into 2a and 2b	Reassess eligibility, Randomization (1:1), First Dose of Study Treatment
Double-Blind Period	Minimum 16 weeks	SC Injections, Monitoring for Asthma Exacerbations, Questionnaires (ACQ-IA)
Open-Label Ext.	48 to 104+ weeks	Continued Benralizumab SC administration for long-term safety and efficacy evaluation
End-of-Treatment	8 weeks post-last dose	Final follow-up assessments after the last dose in the OLE

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to first asthma exacerbation. **Measurement Method:** Clinical assessment evaluating the effect of benralizumab on asthma exacerbations in paediatric and adolescent patients with uncontrolled asthma during the double-blind treatment period compared to placebo.

7.2 Secondary Endpoints * Change from baseline, during the DB treatment period in Interviewer-Administered Version of the Asthma Control Questionnaire (ACQ-IA). * Change from baseline, during the DB treatment period in Asthma symptom score. * Change from baseline, during the DB treatment period in rescue medication use.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring via patient/observer-reported symptom diaries (PASO/Asthma Daily Diary) and clinical visits. **Serious Adverse Events (SAEs):** Routine monitoring for severe asthma exacerbations and potential hypersensitivity/anaphylaxis events. **Data Monitoring Committee (DMC):** Utilized as per standard Phase III regulatory protocols for pediatric populations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy (time-to-first exacerbation), Safety Set for AE/SAE occurrences. **Sample Size Justification:** Target enrollment of $N=200$ patients to provide sufficient statistical power to detect a significant delay in time-to-first exacerbation in the benralizumab arm versus placebo. **Handling of Missing Data:** Standard handling for time-to-event endpoints (e.g., right-censoring for patients who complete the DB period without an exacerbation or who drop out).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of investigator sites to verify the accuracy of the Asthma Daily Diary, questionnaires, and PASO reports. **Audit Trail:** Robust tracking of randomization, eCRF inputs, and placebo-matching blinding procedures.

11. Study Identifiers & Governance

Primary ID: D3250C00024 **Registry Number:** NCT05692180 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Multicentre, Randomised, Double-blind, Parallel Group, Placebo-controlled, Time-to-first Asthma Exacerbation Phase III Efficacy and Safety Study of Benralizumab in Paediatric Patients With Severe Eosinophilic Asthma (DOMINICA) **Official Title:** Efficacy and Safety of Benralizumab in Paediatric Patients With Severe Eosinophilic Asthma (DOMINICA)

12. Clinical Framework (Eosinophilic Asthma Specific)

Disease Area: Asthma **Preferred UMLS Name:** Asthma, severe eosinophilic **Diagnosis Threshold:** Blood eosinophil count ≥ 300 cells/ μL (or $150\text{--}299$ cells/ μL with historical tissue evidence) **Medical Methodology:** Add-on biologic therapy to

high-dose inhaled corticosteroids (ICS) and standard controllers
Optimization Target: Eosinophil depletion to reduce exacerbation frequency

13. Study Procedures & Methodology

Management Pathway: Standard maintenance therapy plus add-on targeted biologic injection (SC) **Education Protocol (HCP):** Administration protocol for pediatric subcutaneous injections
Education Protocol (Patient): Daily symptom tracking and compliance via the Paediatric Asthma Symptom - Observer reported (PASO) or Asthma Daily Diary **Quality Audit Mechanism:** Medication compliance auditing (\$\ge 70\%\$ required during screening)

14. Observation & Follow-Up Schedule

Total Duration: Minimum 16-week DB period, plus up to 104 weeks Open-Label Extension, with an end-of-treatment follow-up 8 weeks after the final dose. **Onsite Visit Cadence:** Regular injection and assessment intervals based on SC dosing schedule **Remote Monitoring Frequency:** Continuous via daily digital/diary tracking **Checklist Review Interval:** Regular review of PASO, ACQ-IA, and exacerbation criteria

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				